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CausalMixGPD: An R Package for Bayesian Nonparametric Conditional Density Modeling in Causal Inference and Clustering with A Modular Heavy-Tail Extension

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Abstract

Heavy-tailed outcomes arise frequently in settings where rare extreme events drive risk, cost, or policy decisions. Parametric models often provide adequate descriptions of the location but can mischaracterize tail behavior, whereas fully nonparametric approaches lack clear mechanisms for tail extrapolation. This article introduces CausalMixGPD, an R package for Bayesian modeling of heavy-tailed outcomes that splices Dirichlet process mixture models for the bulk of the distribution with optional generalized Pareto tails in a single modeling framework. The package accommodates both unconditional and covariate-dependent mixture specifications, performs posterior computation via Markov chain Monte Carlo through nimble, and supports arm-specific outcome models for causal inference. Posterior predictive summaries are provided for densities, quantiles, means, survival curves, and causal contrasts, with particular emphasis on extreme quantiles and other tail-sensitive functionals. The same framework also supports predictor-dependent clustering under multiple kernel families. CausalMixGPD therefore supports heavy-tailed density estimation, model-based clustering, and quantile- and tail-focused causal reporting within a common interface.

Keywords: Bayesian nonparametrics; Dirichlet Process Mixture; Generalized Pareto Distribution; Extreme Value Theory; Cluster Analysis; Causal Inference; R; nimble.

1. Introduction

Causal inference is conventionally summarized through average treatment effects, yet many substantive applications demand full-distribution comparisons between treatment regimes. In clinical trials, for instance, investigators may wish to characterize how a treatment alters

outcomes across the entire outcome scale, including effects among individuals in the upper tail where events are rare but consequences can be severe. This shift from mean-based to distribution-based causal questions ties the credibility of causal analyses directly to the quality of the underlying outcome distribution estimates under competing treatment assignments.

A growing R ecosystem supports Bayesian causal inference through flexible outcome modeling and posterior uncertainty quantification, including tree-based and model-averaging approaches such as bcf, bartCause, and BCEE (R Package Authors 2024a, 2025a, 2023a). While these tools are effective for mean and heterogeneous effect estimation, quantile and tail reporting poses additional challenges: quantiles estimated independently can cross, and tail-focused contrasts require the full outcome distribution rather than a collection of marginal summaries. CausalMixGPD addresses these challenges by modeling outcome densities directly and deriving quantiles, means, survival curves, and tail summaries from a single posterior predictive object.

Dirichlet process mixtures (DPM) offer a versatile nonparametric framework for density estimation, as they can accommodate multimodality and latent heterogeneity without prespecifying the number of mixing components (Escobar and West 1995). Within the R ecosystem, Bayesian nonparametric mixture modeling is well supported by several packages based on Dirichlet process ideas. dirichletprocess provides a flexible interface for specifying DPMs and fitting them by posterior simulation, yielding inference for mixture weights, component labels, and predictive densities under user-chosen kernels (R Package Authors 2023b). DPpackage offers a broad suite of Bayesian semi- and nonparametric models, including DP-based formulations for marginal and conditional densities, regression functions, and model-based predictive distributions (Jara, Hanson, Quintana, Müller, and Rosner 2011). More recently, BNPmix emphasizes practical Bayesian nonparametric mixture approaches, supporting posterior inference for clustering structure and density/regression estimation under DP and Pitman–Yor-type mixture specifications (Corradin, Canale, and Nipoti 2021).

The clustering perspective is also central to the present package. Because DP mixtures induce random partitions through latent allocation variables, they naturally support model-based subgroup discovery in addition to density estimation. Bayesian cluster analysis is typically reported through label-invariant summaries such as pairwise co-clustering probabilities (Binder 1978) and representative partitions selected via Dahl's least-squares criterion (Dahl 2006). In regression settings, clustering itself can depend on predictors: dependent Dirichlet process constructions allow covariates to modulate mixture weights, component-specific parameters, or both (MacEachern 1999, 2000; De Iorio, Müller, Rosner, and MacEachern 2004; Quintana, Müller, Jara, and MacEachern 2022), and predictor-dependent stick-breaking schemes provide a practical route for covariate-modulated cluster prevalence (Ren, Du, Carin, and Dunson 2011). Existing software supports portions of this workflow but typically focuses on bulk density estimation and clustering under a single kernel family without explicit accommodation for heavy-tailed data. Consequently, applications requiring both latent subgroup discovery and tail extrapolation still necessitate additional modeling outside the standard package workflow.

Estimating quantities far into the tail demands special care when data become scarce, and fitting a single model to heavy-tailed data can distort the fit in the central region. Extreme value theory (EVT) provides a principled framework for tail modeling and extrapolation. EVT characterizes the asymptotic distribution of threshold exceedances, enabling estimation of tail probabilities and quantiles beyond the range of observed data (Balkema and de Haan 1974; Pickands 1975; Coles 2001). Bayesian implementations of EVT are available in R for

settings requiring posterior uncertainty quantification of tail quantities. evdbayes provides Bayesian inference for standard extreme-value models via posterior simulation, enabling posterior and posterior-predictive summaries for extreme-event quantities (R Package Authors 2025b). For peaks-over-threshold (POT) approaches, POT implements a focused collection of tools for GPD modeling and threshold-based tail analysis (R Package Authors 2024b). For extreme quantile estimation and rare-event tail summaries, extremefit provides a dedicated software implementation (Durrieu, Grama, Jaunatre, Pham, and Tricot 2018). These existing tools typically emphasize either flexible bulk modeling or dedicated tail analysis; applications requiring both predictor-dependent clustering and causal contrasts in the upper tail still oblige users to integrate multiple disparate workflows.

This article introduces CausalMixGPD, an R package that combines Bayesian nonparametric modeling with EVT, clustering, and causal analysis in the presence of covariates. The package provides three main capabilities:

(i) **BNP conditional density estimation with an optional GPD tail:** The package fits DPMs below a threshold and can splice a generalized Pareto tail above it for tail probabilities, extreme quantiles, and rare-event prediction.

(ii) **Predictor-dependent clustering:** A dedicated clustering framework that models different forms of covariate dependence in latent allocation and reports label-invariant summaries via posterior similarity matrices and Dahl representative partitions (Binder 1978; Dahl 2006).

(iii) **Causal estimation via posterior functionals:** Treatment-specific outcome distributions are modeled directly via conditional density estimation, so that average, restricted-mean, and quantile treatment effects are computed from the fitted distributions in both randomized and observational studies, with optional propensity-score augmentation in the latter.

Overall, CausalMixGPD provides a common interface for density estimation, prediction, clustering, and causal inference when tail behavior is of concern. The package is implemented in R, with computationally intensive components delegated to nimble (R Package Authors 2025c). The remainder of this article is organized as follows. Section 2 introduces the spliced model and the reported causal estimands. Section 3 describes the package workflow and user interface. Sections 4 and 5 illustrate the clustering and causal workflows through data analyses, and Section 6 concludes with a discussion of limitations and future directions.

2. Background and Preliminaries

Let the observation-covariate vector be denoted by

$$(Y, X) \in \mathbb{R} \times \mathbb{R}^p.$$

Then the conditional cumulative distribution function (CDF) and probability density function (PDF), for some covariate x, are defined as:

$$F(y \mid x) = \Pr\,(Y \leq y \mid X = x), \qquad f(y \mid x) = \frac{\partial}{\partial y} F(y \mid x),$$

and the conditional quantile function is given by

$$Q(\tau \mid x) = \inf\Big\{y \in \mathbb{R} :\, F(y \mid x) \geq \tau\Big\}, \qquad 0 < \tau < 1.$$

In CausalMixGPD, $f(\cdot|x)$ is regarded as the inferential target. The central part of the distribution is assumed to be nonparametric, while the upper tail may be parametrized using a generalized Pareto model, when extrapolation to extremes is necessary.

2.1. Generalized Pareto Distribution for Threshold Exceedances

The tail region is defined by values exceeding a high threshold. Extreme value theory provides a standard approximation for such exceedances. Let $u(x)$ denote a high threshold, potentially depending on x, and let the exceedance be given by

$$Z = Y - u(x),$$

considered conditionally on $Y > u(x)$. The conditional excess distribution is given by

$$F_u(z \mid x) = \Pr\big(Y - u(x) \le z \mid Y > u(x), X = x\big), \qquad z \ge 0.$$

Under broad regularity conditions, $F_u(\cdot \mid x)$ is well-approximated by a generalized Pareto distribution (GPD) when $u(x)$ is sufficiently large (Balkema and de Haan 1974; Pickands 1975; Coles 2001).

For $y > u(x)$, the corresponding GPD cumulative distribution function is given by

$$F_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} 1 - \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi}, & \xi \neq 0, \\ 1 - \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0, \end{cases}$$

with support $y \ge u(x)$ when $\xi \ge 0$, and $u(x) \le y \le u(x) - \sigma(x)/\xi$ when $\xi < 0$. The corresponding density function is given by

$$f_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} \dfrac{1}{\sigma(x)} \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi - 1}, & \xi \neq 0, \\ \dfrac{1}{\sigma(x)} \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0. \end{cases}$$

The associated GPD quantile function takes the form

$$Q_{\mathrm{GPD}}(\tau \mid u(x), \sigma(x), \xi) = \begin{cases} u(x) - \sigma(x) \log(1 - \tau), & \xi = 0, \\ u(x) + \dfrac{\sigma(x)}{\xi}\Big((1 - \tau)^{-\xi} - 1\Big), & \xi \neq 0, \end{cases} \qquad 0 < \tau < 1.$$

Figure 1 illustrates the GPD density for several values of the shape parameter ξ with a fixed scale parameter σ and threshold u. The shape parameter governs tail heaviness: $\xi < 0$ yields a light tail with finite upper endpoint, $\xi = 0$ produces an exponential tail, and $\xi > 0$ generates a heavy tail with infinite support. This flexible family for modeling exceedances above a high threshold serves as the tail component in the spliced model described below.

2.2. Dirichlet Process Mixture Model for Conditional Densities

Below the threshold $u(x)$, the conditional density of Y given X is modeled using a DPM regression model. Let $k(\cdot \mid x; \theta)$ denote a user-chosen kernel family indexed by parameter θ. The hierarchical specification is given by

$$Y \mid X = x, \theta \sim k(\cdot \mid x; \theta), \qquad \theta \mid H \sim H, \qquad H \mid \kappa, H_0 \sim \mathrm{DP}(\kappa, H_0),$$

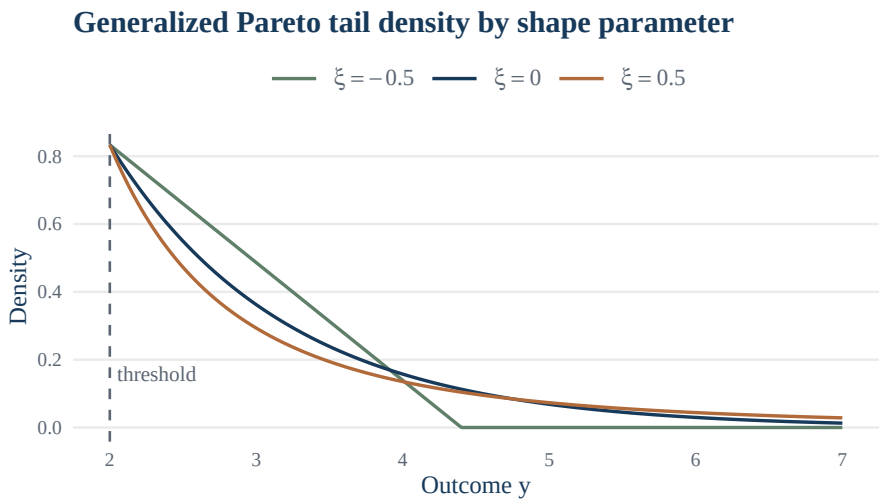


Figure 1: Generalized Pareto distribution (GPD) density above a fixed threshold $u = 2$ with scale $\sigma = 1.2$ and three values of the shape parameter, $\xi \in \{-0.5, 0, 0.5\}$. The dashed vertical line marks the threshold u.

where H is an unknown mixing distribution on the kernel-parameter space, H_0 is a base measure, and $\kappa > 0$ is a concentration parameter (Ferguson 1973). Marginalizing over H yields an (almost surely) discrete mixing distribution together with an infinite mixture representation of the form

$$f_{\mathrm{DP}}(y \mid x) = \int k(y \mid x; \theta)\, dH(\theta) = \sum_{j=1}^{\infty} w_j\, k(y \mid x; \theta_j), \qquad w_j \geq 0, \qquad \sum_{j=1}^{\infty} w_j = 1.$$

One representation of the weights is Sethuraman's stick-breaking construction (Sethuraman 1994), in which

$$w_1 = V_1, \qquad w_j = V_j \prod_{\ell<j} (1 - V_\ell), \qquad V_j \overset{\text{iid}}{\sim} \mathrm{Beta}(1, \kappa), \qquad \theta_j \overset{\text{iid}}{\sim} H_0. \tag{1}$$

An equivalent characterization is obtained by marginalizing over H, yielding the Blackwell–MacQueen Pólya urn representation. In particular, the conditional distribution of a parameter value given all others can be written as

$$\theta_i \mid \{\theta_j\}_{j \neq i} \sim \frac{\kappa}{\kappa + n} H_0(\cdot) + \frac{1}{\kappa + n} \sum_{j=1}^{n} \delta_{\theta_j}(\cdot),$$

where $\delta_{\theta_j}(\cdot)$ denotes a point mass at θ_j. This specification induces a clustering structure commonly described via the Chinese restaurant process (CRP) (Neal 2000). Both the stick-breaking and CRP representations are widely employed for posterior simulation in DPM models (Escobar and West 1995; Neal 2000). Because DPMs operate without prespecifying the number of mixture components and can capture complex density features such as multimodality, skewness, and heteroscedasticity through appropriate kernel choices, they constitute a natural model for the bulk of the conditional distribution where data are abundant and complex structure may be present.

2.3. Spliced DPM–GPD Conditional Distribution

CausalMixGPD employs a spliced (bulk–tail) construction that retains the DPM fit below a high threshold and replaces the upper tail with a GPD component (Frigessi, Haug, and Rue

2002; Behrens, Lopes, and Gamerman 2004; Coles 2001). For a fixed covariate value x, let $u(x)$ denote the threshold, $\sigma(x) > 0$ the tail scale above that threshold, and $\xi \in \mathbb{R}$ the tail-shape parameter. Let Θ collect the bulk-mixture parameters and Φ collect the tail quantities $\{u(x), \sigma(x), \xi\}$. Write $F_{\mathrm{DP}}(y \mid x; \Theta)$ for the bulk conditional CDF and $F_{\mathrm{GPD}}(y \mid x; \Phi)$ for the exceedance CDF. The fitted bulk mass below the threshold $u(x)$ is given by

$$p_u(x) = F_{\mathrm{DP}}(u(x) \mid x; \Theta),$$

and the spliced CDF is given by

$$F(y \mid x; \Theta, \Phi) = \begin{cases} F_{\mathrm{DP}}(y \mid x; \Theta), & y \le u(x), \\ p_u(x; \Theta) + \{1 - p_u(x; \Theta)\}\, F_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{2}$$

which is continuous at $u(x)$ by construction. Differentiating (2), the spliced density is given by

$$f(y \mid x; \Theta, \Phi) = \begin{cases} f_{\mathrm{DP}}(y \mid x; \Theta), & y \le u(x), \\ \{1 - p_u(x; \Theta)\}\, f_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{3}$$

where the tail density is rescaled by $1 - p_u(x; \Theta)$ to ensure that $f(\cdot \mid x; \Theta, \Phi)$ integrates to unity. Figure 2 provides a visual comparison of a one-component gamma density, a two-component gamma-mixture density, and a two-component gamma-mixture-plus-GPD spliced density with the splice point set at the 80th bulk-mixture quantile.

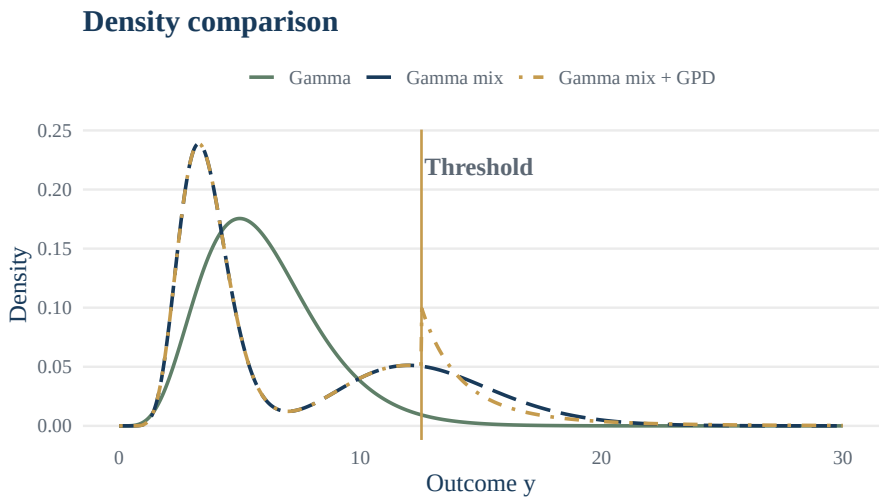


Figure 2: Three density curves on the same axes: a single-component gamma density (solid), a two-component gamma mixture density (long-dash), and a spliced gamma-mixture-plus-GPD density (dot-dash) with GPD tail scale 2.0 and shape 0.25. The vertical line marks the splice threshold, set at the 80th percentile of the two-component gamma bulk mixture.

For $\tau \le p_u(x; \Theta)$, quantiles are obtained by inverting the bulk CDF. Let $Q_{\mathrm{DP}}(\tau \mid x; \Theta)$ denote the bulk quantile function. For $\tau > p_u(x; \Theta)$, the probability index must be rescaled relative to the exceedance probability, yielding the rescaled index

$$\tilde{\tau}(x; \Theta) = \frac{\tau - p_u(x; \Theta)}{1 - p_u(x; \Theta)} \in (0, 1),$$

Combining both cases, the spliced quantile function is given by

$$Q(\tau \mid x; \Theta, \Phi) = \begin{cases} Q_{\mathrm{DP}}(\tau \mid x; \Theta), & \tau \le p_u(x; \Theta), \\ Q_{\mathrm{GPD}}(\tilde{\tau}(x; \Theta) \mid x; \Phi), & \tau > p_u(x; \Theta). \end{cases} \tag{4}$$

When the bulk kernel possesses a finite mean and the GPD tail satisfies $\xi < 1$, the mean of the spliced model admits the closed form

$$\mathrm{E}(Y \mid x; \Theta, \Phi) = \int_{-\infty}^{u(x)} y\, f_{\mathrm{DP}}(y \mid x; \Theta)\, dy \;+\; \{1 - p_u(x; \Theta)\} \left\{ u(x) + \frac{\sigma(x)}{1-\xi} \right\}.$$

This identity underlies the analytical mean calculations employed by the package whenever the mean exists.

2.4. Clustering Under Bayesian Nonparametric Mixtures

The modeling scheme within the Bayesian nonparametric framework starts with representing the set $\{f(\cdot \mid x) : x \in \mathcal{X}\}$ using predictor-indexed random probability measures. Within the Dependent Dirichlet Process setting of MacEachern (1999, 2000), one assumes a collection $\{G_x : x \in \mathcal{X}\}$ of random mixing measures whose dependence allows borrowing of information in $f(\cdot \mid x)$.

In general terms, the model takes the form

$$f(y \mid x) \;=\; \int k(y \mid x, \theta)\; dG_x(\theta),$$

where $k(\cdot \mid x, \theta)$ denotes a kernel, and G_x is a predictor dependent random mixing measure. If a mixture measure G_x is discrete (e.g., constructed via stick-breaking), latent parameters exhibit ties. This induces partitioning of observations via a latent variable representation with cluster labels z_i, and component parameters $\{\theta_j\}_{j \geq 1}$ defined by

$$y_i \mid z_i, \{\theta_j\}, x_i \sim k(\cdot \mid x_i, \theta_{z_i}).$$

Clustering is thus encoded in the induced partition $\Pi = \{z_1, \ldots, z_n\}$'s posterior distribution and is characterized by label-invariant co-clustering statistics (Binder 1978; Dahl 2006). The clustering capabilities of CausalMixGPD incorporate three types of dependence in predictors, each corresponding to specific mixing measure constructions G_x.

- **Weight dependence only:** Dependence enters through predictor-modulated mixture weights, whereas component parameters are shared across predictors,

  $$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid \theta_j).$$

  One implementation uses predictor-dependent stick-breaking (MacEachern 2000; Quintana *et al.* 2022). In particular, stick lengths become regression functions of predictors, with logistic sticks corresponding to logistic regressions mapping to weights (Ren *et al.* 2011). These models are instances of the DDP framework, in which dependence is introduced through smooth weight variation.

- **Parameter dependence only:** Here, dependence arises through predictor-modulated component parameters, while weights remain constant,

  $$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j\, k(y \mid x, \theta_j).$$

  This type of dependence matches the DDP framework with shared weights but predictor-indexed component-specific kernel parameters (MacEachern 1999, 2000). This class of models resembles dependent ANOVA prior constructions of random measures (De Iorio *et al.* 2004).

- **Weight and parameter dependence:** The most flexible specification allows predictors to influence both cluster prevalence and within-cluster behavior,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid x, \theta_j)\,.$$

  This corresponds to DDP-type models in which both selection probabilities and component-specific conditional structure vary with predictors (Quintana *et al.* 2022).

Posterior inference and prediction for cluster allocation. Let $D_n = \{(y_i, x_i)\}_{i=1}^n$ denote the observed data and let $z_i \in \{1, \ldots, J\}$ be the latent component label under a finite truncation of the mixture model. At MCMC iteration m, let $\Theta^{(m)} = \{\theta_j^{(m)}\}_{j=1}^J$ denote the component-specific parameters and $w_j^{(m)}(x)$ the predictor-dependent mixture weights. The corresponding conditional density is

$$f^{(m)}(y \mid x) = \sum_{j=1}^{J} w_j^{(m)}(x)\, k(y \mid x, \theta_j^{(m)}),$$

with $w_j^{(m)}(x) \equiv w_j^{(m)}$ in the predictor-independent case.
For each unit i, the conditional posterior probability of allocation to component j at draw m is

$$p_{ij}^{(m)} = \frac{w_j^{(m)}(x_i)\, k(y_i \mid x_i, \theta_j^{(m)})}{\sum_{\ell=1}^{J} w_\ell^{(m)}(x_i)\, k(y_i \mid x_i, \theta_\ell^{(m)})},$$

and we estimate it by Monte Carlo averaging,

$$\widehat{p}_{ij} = \frac{1}{M} \sum_{m=1}^{M} p_{ij}^{(m)}.$$

Because mixture labels are only identifiable up to permutation, we report clustering summaries through label-invariant posterior functionals. In particular, we estimate the posterior similarity matrix (PSM) as

$$\widehat{S}_{i\ell} = \frac{1}{M} \sum_{m=1}^{M} \mathbb{1}\{z_i^{(m)} = z_\ell^{(m)}\}, \qquad i, \ell = 1, \ldots, n,$$

which approximates $\Pr(z_i = z_\ell \mid \text{data})$. A representative partition is then obtained via Dahl's least-squares method, i.e., by selecting the sampled partition whose adjacency matrix is closest to $\widehat{S}$ under squared loss (Dahl 2006). We therefore use the PSM as the main uncertainty summary, and the representative partition as a stable point estimate; further implementation details are given in Section 3.3.
If a hard allocation is needed for the training sample, observation i may be assigned to

$$\widehat{z}_i^{\text{train}} = \arg\max_{1 \le j \le J} \widehat{p}_{ij},$$

though substantive interpretation relies on the representative partition rather than the raw component indices.

For a new covariate profile $x^\star$, if the response is unobserved then the posterior predictive allocation probabilities reduce to the weights,

$$p_j^{(m)}(x^\star) = w_j^{(m)}(x^\star), \qquad \widehat{p}_j(x^\star) = \frac{1}{M}\sum_{m=1}^{M} w_j^{(m)}(x^\star).$$

If $y^\star$ is also observed, the posterior allocation probabilities are

$$p_j^{(m)}(x^\star, y^\star) = \frac{w_j^{(m)}(x^\star)\, k(y^\star \mid x^\star, \theta_j^{(m)})}{\sum_{\ell=1}^{J} w_\ell^{(m)}(x^\star)\, k(y^\star \mid x^\star, \theta_\ell^{(m)})}, \qquad \widehat{p}_j(x^\star, y^\star) = \frac{1}{M}\sum_{m=1}^{M} p_j^{(m)}(x^\star, y^\star).$$

In the package output, predictions for new observations are ultimately expressed relative to the representative training partition. Thus, in-sample inference is summarized by the PSM and its representative partition, while out-of-sample inference is summarized by posterior assignment scores and the induced allocation to the representative training clusters.

2.5. Causal Inference: Notation, Identification, and Reported Estimands

In this package, we use the potential outcomes framework for causal inference introduced by Rubin (1974). In particular, $A \in \{0, 1\}$ is an indicator for a binary treatment assignment ($A = 1$ treated, $A = 0$ control), X denotes pre-treatment covariates, and $Y(a)$ denotes the potential outcome under treatment arm $a \in \{0, 1\}$. The observed outcome is then defined as

$$Y = A\,Y(1) + (1 - A)\,Y(0).$$

For each arm $a \in \{0, 1\}$, the arm-specific conditional CDF can be expressed as

$$F_a(y \mid x) = \Pr\{Y(a) \le y \mid X = x\}, \qquad a \in \{0, 1\}.$$

Under the CausalMixGPD approach, the arm-specific conditional densities $f_a(y \mid x)$ follow the spliced DP mixture form given in (3). The conditional τ-quantile function can be computed by taking the inverse CDF,

$$Q_a(\tau \mid x) = \inf\{y : F_a(y \mid x) \ge \tau\}, \qquad \tau \in (0, 1),$$

whereas the conditional mean is computed via integration of the conditional density function,

$$\mu_a(x) = \mathrm{E}\{Y(a) \mid X = x\} = \int y\, f_a(y \mid x)\, dy.$$

Marginal distributions are obtained by averaging conditional distributions over a covariate distribution. Writing $F_a^{\mathrm{m}}(\cdot)$ for the population arm-a CDF and $F_a^{\mathrm{t}}(\cdot)$ for the treated analog, these are given by

$$F_a^{\mathrm{m}}(y) = \mathrm{E}\{F_a(y \mid X)\}, \qquad F_a^{\mathrm{t}}(y) = \mathrm{E}\{F_a(y \mid X) \mid A = 1\}.$$

The associated marginal quantile functions are obtained by inverting these standardized CDFs,

$$Q_a^{\mathrm{m}}(\tau) = \inf\{y : F_a^{\mathrm{m}}(y) \ge \tau\}, \qquad Q_a^{\mathrm{t}}(\tau) = \inf\{y : F_a^{\mathrm{t}}(y) \ge \tau\},$$

while the marginal means integrate the conditional mean over the respective covariate distributions,

$$\mu_a^{\mathrm{m}} = \mathrm{E}\{\mu_a(X)\}, \qquad \mu_a^{\mathrm{t}} = \mathrm{E}\{\mu_a(X) \mid A = 1\}.$$

The identification of such estimands depends on the study design. For randomized experiments, randomness and positivity result in exchangeability between treatment assignment and potential outcomes (Imbens and Rubin 2015). For observational studies, the conditions for identification include the following conditions (Rosenbaum and Rubin 1983; Imbens and Rubin 2015).

- **Stable Unit Treatment Value Assumption (SUTVA):** there is no interference between units and no hidden versions of treatment.
- **Consistency:** if $A = a$, then the observed outcome equals the corresponding potential outcome, that is, $Y = Y(a)$.
- **Conditional ignorability:** treatment assignment is as-if randomized given covariates, so that
$$\big(Y(0), Y(1)\big) \;\perp\; A \;\mid\; X.$$
- **Overlap (positivity):** every covariate profile has a strictly positive probability of receiving either treatment,
$$0 < \Pr\,(A = 1 \mid X = x) < 1$$
for all covariate values of interest.

Under these conditions, the treatment-specific conditional distributions can be inferred based on the observations as follows:

$$F_a(y \mid x) = \Pr\,(Y \le y \mid A = a, X = x), \qquad a \in \{0, 1\}.$$

To address the potential unobserved confounding in the observational setting, the package uses the propensity score (PS) method during the design phase (Rosenbaum and Rubin 1983). The PS is the conditional probability of receiving the treatment given the covariates:

$$\rho(x) = \Pr\,(A = 1 \mid X = x).$$

A key property of the propensity score is that conditioning on it balances the covariates in both treatment groups. Following a two-step procedure, the package first obtains an estimate of $\rho(x)$ using the data (A, X) alone and then adds the estimated propensity score to the outcome model covariates. Thus, the augmented predictor vector will take the following form:

$$r(x) = \big(x^{\mathsf{T}}, \widehat{\rho}(x)\big)^{\mathsf{T}}.$$

Next, the conditional distributions of the outcome are estimated separately for each treatment group using the augmented predictor $r(x)$, and causal contrasts are evaluated as functionals of the treatment-specific distributions. Figure 3 depicts two treatment-specific conditional density curves at a representative covariate value. The dashed vertical lines correspond to the 90th conditional quantile for each arm, while the double-ended arrow shows the conditional quantile treatment effect (CQTE).

The package reports the causal estimands listed in Table 1, where superscripts m and t denote full-population and treated-population standardization, respectively.

The next section describes how the package fits the treatment-specific conditional distributions and computes these estimands from posterior predictive output.

3. Package Overview: Workflow, Model, and Inference

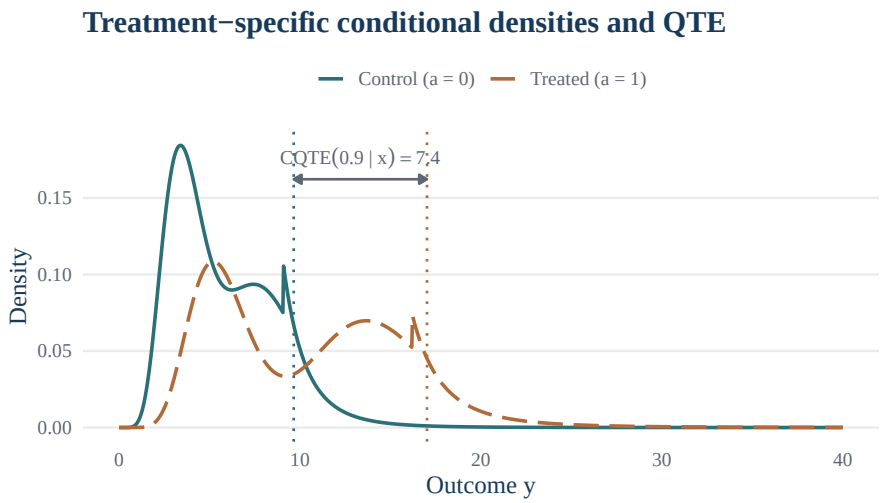


Figure 3: Two arm-specific conditional outcome densities $f_a(y \mid x)$, $a \in \{0, 1\}$, at a representative covariate profile x under a spliced gamma-mixture-plus-GPD model: control arm ($a = 0$, solid) and treated arm ($a = 1$, long-dash). Dotted vertical lines mark the 90th conditional quantiles $Q_0(0.9 \mid x)$ and $Q_1(0.9 \mid x)$, and the double-headed arrow connects them, labelled with the conditional quantile treatment effect $\mathrm{CQTE}(0.9 \mid x) = Q_1(0.9 \mid x) - Q_0(0.9 \mid x)$.

Table 1: Causal estimands reported by CausalMixGPD. Columns list the estimand name, its defining contrast as a functional of the treatment-specific outcome distributions, and the target population.

Estimand	Formula	Target population
ATE	$\mu_1^{\mathrm{m}} - \mu_0^{\mathrm{m}}$	Entire population
ATT	$\mu_1^{\mathrm{t}} - \mu_0^{\mathrm{t}}$	Treated sub-population
QTE(τ)	$Q_1^{\mathrm{m}}(\tau) - Q_0^{\mathrm{m}}(\tau)$	Entire population at index τ
QTT(τ)	$Q_1^{\mathrm{t}}(\tau) - Q_0^{\mathrm{t}}(\tau)$	Treated sub-population at index τ
CATE(x)	$\mu_1(x) - \mu_0(x)$	Individual with covariate profile x
CQTE($\tau \mid x$)	$Q_1(\tau \mid x) - Q_0(\tau \mid x)$	Individual with x at index τ

This section describes the workflow for fitting conditional models and obtaining predictions with CausalMixGPD. The presentation begins with the one-arm setup, in which the spliced DPM–GPD model is fitted to an outcome conditional on covariates. The causal extension augments this framework with arm-specific outcome models and an optional propensity score augmentation for observational studies. Simulated data sets bundled with the package are used throughout to illustrate the main features; these are accessible via `data()`.

3.1. Spliced Regression model, Likelihood, and Covariate Dependence

The observed data consist of n outcome–covariate pairs $\mathcal{D} = \{(y_i, x_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^p$ denotes the covariate vector for unit i, optionally including an intercept column.

```
data("nc_posX100_p3_k2", package = "CausalMixGPD")
dat <- data.frame(y = nc_posX100_p3_k2$y, nc_posX100_p3_k2$X)
```

The conditional target distribution $F(\cdot \mid x)$ and the DPM–GPD spliced construction are provided in Section 2, with the resulting spliced CDF and density functions given in (2)–(3). To perform posterior inference, CausalMixGPD truncates the infinite mixture into a finite DPM with J components such that

$$f_{\mathrm{DP}}(y \mid x; \Theta) \approx \sum_{j=1}^{J} w_j \, k(y \mid x; \theta_j),$$

where $j = 1, \ldots, J$ indexes each mixture component, $w_j \geq 0$ are the corresponding mixture component weights, and the sum equals one up to some residual probability mass allocated to other components. Kernel parameters can be constant, specific to a mixture component, or dependent on the covariates according to link functions of the form

$$g(\psi_j(x)) = x^\top \beta_{j,\psi},$$

Here, $\psi_j(x)$ is a generic kernel parameter for component j with input covariates x, while the link function $g(\cdot)$ restricts the parameter support. All code illustrations in this section use the MCMC sampling options shown below; longer sampling and multiple chains are encouraged in applied settings.

```
data.frame(mcmc_fixed,row.names = NULL)


  niter nburnin thin nchains seed
1  2000     500    1       1 2026
```

The single-arm splicing approach can be performed by passing a formula, data, and control options to `dpmgpd()`. The `formula` argument uses standard R syntax to indicate which covariates enter both the bulk and tail DPM components. The `backend` option indicates whether the DPM component weights should be represented using stick breaking in (1) or the Chinese restaurant process formulation. The `kernel` argument selects the bulk mixture kernel, `components` controls the number of components, and `mcmc` sets sampling

```
fit <- dpmgpd(formula = y ~ x1 + x2 + x3, data = dat, mcmc = mcmc_fixed)
```

Given the spliced density in (3), the likelihood is given by

$$L(\Theta, \Phi; \mathcal{D}) = \prod_{i=1}^{n} f(y_i \mid x_i; \Theta, \Phi), \tag{5}$$

where Θ collects the bulk parameters $\{w_j, \theta_j\}_{j=1}^{J}$ and Φ collects the tail parameters governing $u(x)$, $\sigma(x)$, and ξ. The log-likelihood can equivalently be written as

$$\begin{aligned} \log L(\Theta, \Phi; \mathcal{D}) \;=\; & \sum_{i=1}^{n} (1 - \delta_i) \log f_{\mathrm{DP}}(y_i \mid x_i; \Theta) \\ & + \sum_{i=1}^{n} \delta_i \Big[\log\{1 - F_{\mathrm{DP}}(u(x_i) \mid x_i; \Theta)\} + \log f_{\mathrm{GPD}}(y_i \mid x_i; \Phi) \Big], \end{aligned}$$

where f_{GPD} is the GPD tail density from Section 2.3 and $F_{\mathrm{DP}}(\cdot \mid x)$ is the bulk mixture CDF. When the GPD tail is not included, the model reduces to $f(y \mid x; \Theta, \Phi) \equiv f_{\mathrm{DP}}(y \mid x; \Theta)$, a mixture over the bulk alone. A convenience interface to this bulk-only model is provided by the `dpmix()` wrapper.

```
fit <- dpmix(formula = y ~ x1 + x2 + x3, data = dat, mcmc = mcmc_fixed)
```

Prior distributions for Θ and Φ can be specified by the user; defaults are weakly informative. For the DPM parameters Θ, the default prior employs stick-breaking priors for the mixture weights and conjugate priors for kernel parameters when conjugacy obtains. For the tail

parameters Φ, the default adopts a regression-based specification for the threshold $u(x)$ and scale $\sigma(x)$, together with a weakly informative prior for the shape parameter ξ.

3.2. Statistical Inference: Estimation and prediction

Posterior simulation targets the joint posterior, which is given by

$$p(\Theta, \Phi \mid \mathcal{D}) \propto L(\Theta, \Phi; \mathcal{D})\, p(\Theta, \Phi)$$

using the likelihood (5) and the prior structure described in Section 3.1. CausalMixGPD conducts MCMC via nimble (R Package Authors 2025c). The fitted object supports `summary()` for posterior summaries, `plot()` for standard MCMC diagnostics, and `params()` for lightweight parameter extraction.

```
summary(fit)  # posterior summaries for parameters and functionals
params(fit)   # extract MCMC samples for parameters and functionals
plot(fit, family = "traceplot", params = "alpha") # traceplot for concentration parameter
plot(fit, family = "auto") # automatic selection of diagnostic plots for all parameters
```

The package provides posterior summaries for several functionals of the fitted conditional distribution defined in Section 2: the density $f(y \mid x)$, survival function $S(y \mid x) = 1 - F(y \mid x)$, quantile function $Q(\tau \mid x)$, and mean $\mathrm{E}(y \mid x)$. For the `density`, `survival`, and `quantile` types, each functional is evaluated draw-by-draw from (2)–(4), and the retained draws are then summarized. For `type="mean"`, the package employs analytical draw-level calculations whenever the kernel mean exists and, for spliced GPD models, the tail-shape parameter satisfies $\xi < 1$; if these conditions are not met, the ordinary mean is not reported and the user is directed to `type="rmean"`. The restricted-mean target (`type="rmean"`) provides a separate posterior predictive Monte Carlo calculation for a user-supplied cutoff and remains well-defined even when the ordinary mean does not exist. The following code illustrates how posterior density, quantile, and mean summaries are obtained at observed covariate values, together with credible intervals for quantiles and HPD intervals for means.

```
predict(fit, type = "density", interval = "credible", level = 0.95)
predict(fit, type = "quantile", index = c(0.25, 0.5, 0.75))
predict(fit, type = "mean", interval = "hpd", level = 0.9)
```

Predictions at new covariate values are obtained by calling `predict()` with a data set or covariate profiles sharing the same structure as the training data.

```
x_new <- as.data.frame(lapply(dat[, c("x1", "x2", "x3")], quantile,
                              probs = c(0.25, 0.50, 0.75), na.rm = TRUE))
y_grid <- seq(0, 10, length.out = 200)
```

For a covariate profile $\boldsymbol{x}^\star$, predictions are obtained from the posterior predictive density

$$p(y^\star \mid \boldsymbol{x}^\star, \mathcal{D}) = \int f(y^\star \mid \boldsymbol{x}^\star; \Theta, \Phi)\, p(\Theta, \Phi \mid \mathcal{D})\, d\Theta\, d\Phi.$$

Predictions of the density and survival functions involve a grid of y values provided through the `y` argument, while predictions of quantiles require only the probability indices through the `index` argument. The code example that follows provides posterior survival and quantile predictions for new covariate profiles derived from the empirical quartiles of the predictors in the training data.

The argument-level controls for the above methods are provided in Appendix A. Specifically, Section A.1 extends the prior and backend definitions discussed previously, whereas Section A.2 summarizes the full prediction, interval, and diagnostic interfaces.

```
predict(fit, newdata = x_new, y = y_grid, type = "survival", level = 0.95)
predict(fit, newdata = x_new, type = "quantile", index = c(0.5, 0.99), interval = "credible")
```

3.3. Clustering extension

```
data("nc_realX100_p3_k2", package = "CausalMixGPD")
dat_cl <- data.frame(y = nc_realX100_p3_k2$y[1:100], nc_realX100_p3_k2$X[1:100, ])
```

Under the clustering approach, the fitted mixture is regarded as a latent partition, and the output is cluster summaries. For a mixture model without the optional GPD tail, the function `dpmix.cluster()` is called to perform clustering; otherwise, `dpmgpd.cluster()` is used, with predictor dependence introduced via weights, mixture parameters, or both. Since labels assigned to components are arbitrary, the primary results include label-free summaries of the fitted mixture, namely the posterior similarity matrix (PSM), the representative partition, the number of clusters, and the cluster assignment scores (Binder 1978; Dahl 2006). The same prediction interface `predict()` can produce cluster labels for training observations or assign clusters to new observations.

The code block below demonstrates how to fit a clustering model with covariates affecting mixture parameters, by setting `type="param"`.

```
fit_cluster <- dpmix.cluster(y ~ x1 + x2 + x3, data = dat_cl[1:100,], kernel = "laplace",
                                 type = "both", components = 10, mcmc = mcmc_fixed)
```

The function `predict()` accepts two types of predictions: posterior similarity matrix or cluster labels, determined by the `type` argument.

```
predict(fit_cluster, type = "psm")
```

Cluster labels of new observations, along with corresponding assignment scores, are calculated as follows.

```
predict(fit_cluster, newdata = dat_cl[101:110,], type = "label")
```

In Section 2.4, we mentioned that the software supports three types of dependence and evaluates clustering performance based on PSM summaries and representative partitions. Clustering-specific arguments and plotting options are described in Appendix A.3.

3.4. Causal extension and propensity score augmentation

The causal inference workflow consists of estimating the causal framework separately per treatment arm and optionally augmenting the outcome covariates with a summary of the propensity score (PS). In the latter case (e.g., observational study), treatment allocation is assumed to be Bernoulli distributed, as follows,

$$A_i \mid \boldsymbol{x}_i \sim \text{Bernoulli}(e_\gamma(\boldsymbol{x}_i)),$$

and the predicted PS is included into the covariate vector to form an augmented regressor,

$$\boldsymbol{r}(\boldsymbol{x}) = \left(\boldsymbol{x}^\top,\ \psi(\widehat{e}(\boldsymbol{x}))\right)^\top,$$

where the PS link function (logit or probit) is determined by the `PS` parameter and the transformation $\psi(\cdot)$ used on the estimated propensity score depends on `ps_scale`; the default is the identity. Then, the outcome model in each arm is specified as a spliced DPM-GPD regression in the augmented covariates; the structure of splicing mixture regression is identical to that of Section 3.1. In the setting of randomized experiment, no PS estimation is needed, and the causal inference model for each arm uses the exact same modeling workflow as in the one-arm problem.

The code chunk below loads the data set generated from a benchmark setup and constructs the input `data.frame` required by the causal estimation workflows, consisting of the outcome, treatment variable, and covariates.

The invocation of `dpmgpd.causal()` trains spliced mixture regression for each treatment arm, with the causal model having the same structure as `dpmgpd()` but taking into account additional parameters controlling the PS model (`PS`, `ps_scale`, `ps_summary`) and different MCMC sampling strategies for the outcome and PS components (`mcmc_outcome`, `mcmc_ps`).

```
causal_fit <- dpmgpd.causal(formula = y ~ x1 + x2 + x3, data = causal_df, treat = "A",
                            backend = "crp", kernel = "laplace", components = 5,
                            PS = "logit", ps_scale = "logit", ps_summary = "mean",
                            mcmc_outcome = mcmc_fixed, mcmc_ps = mcmc_fixed)
```

Estimated population-averaged quantities are computed using the functions `ate()` and `qte()`, and counterparts with treatment-population averaging are implemented through `att()` and `qtt()`. Confidence intervals are supported through interval estimation, and results can be visualized by invoking the generic `plot()` function with `type="effect"`, which plots estimated effects across quantiles along with the corresponding uncertainty bounds. The below code snippet shows the use of `qtt()` to compute treated population effects.

```
qtt <- qtt(causal_fit, probs = c(0.50, 0.90, 0.95), interval = "credible")
summary(qtt)
plot(qtt, type = "effect")
```

Quantified conditional causal effects based on user-provided covariate profiles can be evaluated by the generic `predict()` function, which returns the selected summary of the posterior predictive outcome distribution depending on the value of the `type` parameter; the examples below produce survival probabilities at a grid of user-defined covariate values covering the empirical quartiles of the training variables.

```
predict(causal_fit, newdata = causal_xgrid, type = "survival", y = rep(4, nrow(causal_xgrid)))
```

The corresponding bulk-only version is `dpmix.causal()`, where the causal estimand and prediction remain unchanged without the GPD tail. For randomized trials, the procedure is identical but without the PS adjustment, where `PS` can be set to `FALSE`. Other causal estimands, PS adjustments, and plots are documented in Appendix A.3.

4. Data Analysis I: Clustering

This section demonstrates clustering under a bulk-only conditional mixture model. The goal is to discover latent subgroups among the observational units using the fitted mixture model, while accounting for partition uncertainty in a label-invariant manner. Unlike the density prediction focus of Section 3, here the interest lies in partition summaries and cluster membership for out-of-sample data. Key outputs include the posterior similarity matrix (PSM), a representative partition, and cluster assignments for held-out observations.

4.1. Data and preprocessing

The Boston data set is used in this analysis, where each observation corresponds to a census tract. The response variable used is medv, representing median home prices in each area. Predictors used include the lstat, rm, and nox variables, which represent the level of neighborhood disadvantage, the size of the houses, and pollution levels, respectively. This combination of predictors constitutes a well-motivated choice for discovering representative clusters. We are interested in uncovering representative clusters from amongst the data, and predicting which latent clusters held-out census tracts belong to using out-of-sample prediction. An 80%/20% train-test split was conducted.

```
library(CausalMixGPD)
library(MASS)

set.seed(123)
data("Boston", package = "MASS")

dat <- Boston
dat <- subset(dat, select = c(medv, lstat, rm, nox))

n <- nrow(dat)
idx_train <- sample(seq_len(n), size = floor(0.80 * n), replace = FALSE)
train_dat <- dat[idx_train, ]
test_dat  <- dat[-idx_train, ]
```

Figure 4 summarizes the distribution of the Boston housing response, median home value (`medv`). The variability in the response is substantial, and the distribution exhibits a pronounced upper tail. These features motivate a flexible distributional model rather than relying only on a simple mean-regression specification.

4.2. Model estimation and training cluster summaries

A bulk only conditional mixture model is applied to the `medv` dataset with three covariates. When `type="weights"`, covariates determine the weights of the mixture while the parameters of the components are common across all covariates. This method is helpful for clustering where the prevalence of latent clusters may differ between neighborhoods.

```
fit_clust <- dpmix.cluster( formula = medv ~ lstat + rm + nox,
                            data  = train_dat, kernel   = "normal",
                             type = "weights", components = 10, mcmc = mcmc_fixed)
```

Since mixture components cannot be identified up to permutation, clusters in the training data are represented using the posterior similarity matrix and sample training labels. The posterior similarity matrix gives the main label-invariant uncertainty summary by recording posterior co-clustering probabilities.

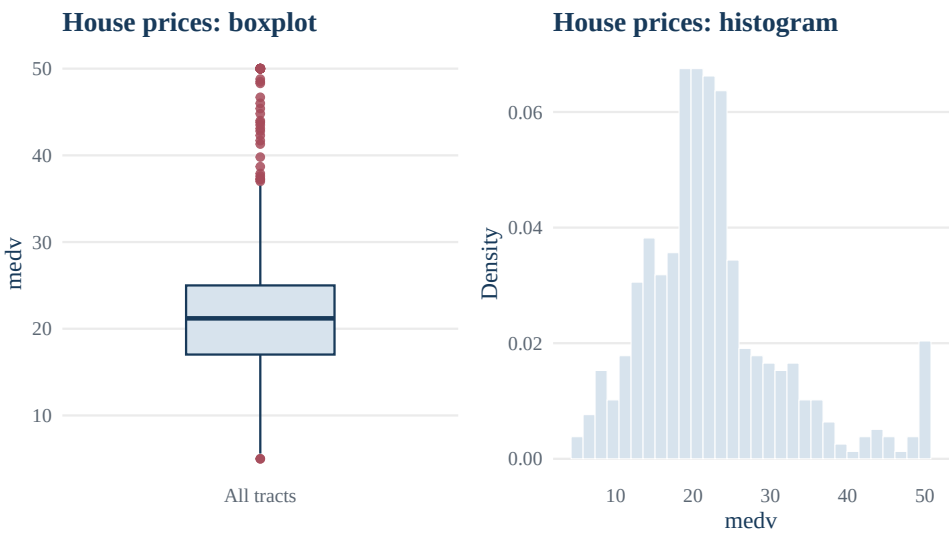


Figure 4: Full-sample distribution of median house values (`medv`, in thousands of US dollars) for the 506 census tracts in the `MASS::Boston` data set. Left panel: boxplot with upper outliers highlighted. Right panel: density-scaled histogram on the same variable.

```
z_train_psm <- predict(fit_clust, type = "psm")
```

```
plot(z_train_psm, type = "summary")
```

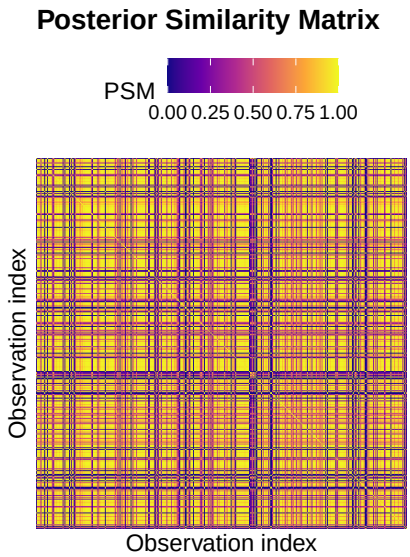


Figure 5: Posterior similarity matrix (PSM) summary for the Boston training sample, returned by `plot(predict(fit_clust,type="psm"),type="summary")`. Each cell (i, ℓ) encodes the posterior co-clustering probability $\Pr(z_i = z_\ell \mid \text{data})$, with rows and columns ordered to expose block structure. Darker shading corresponds to higher co-clustering probability.

As seen from Figure 5, the PSM reveals clear block structure along the diagonal, implying that the model can detect multiple sets of tracts whose co-clustering probability is relatively high. While there are clear separations, there are still some uncertainties for those tracts on the edge of two or more neighboring groups, which reflects reasonable posterior uncertainty in a non-experimental housing dataset, since features of neighborhoods tend to change gradually rather than abruptly.

```
z_train_lab <- predict(fit_clust, type = "label")
```

Figure 6 shows the representative training cluster summary produced by the package `plot()` method applied to the training labels object.

```
plot(z_train_lab, type = "summary")
```

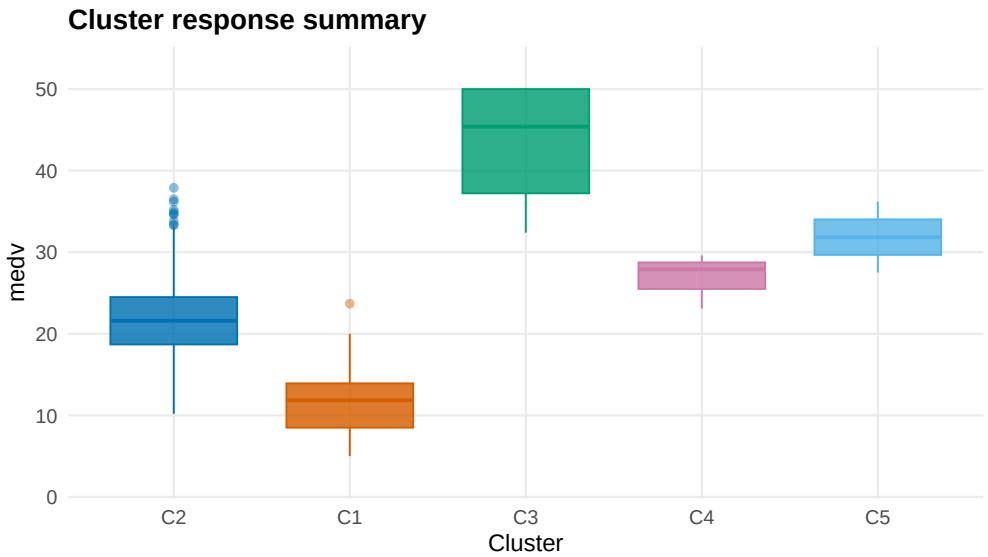


Figure 6: Representative training-cluster summary for the Boston `medv` analysis, returned by the `plot()` method applied to the labels object from `predict(fit_clust,type="label")`. Training observations are shown on the response scale and coloured by their assigned cluster in the representative partition.

The representative partition translates the posterior co-clustering structure into an interpretable summary on the response scale. The inferred groups separate tracts primarily by housing-value level, but they also reflect broader differences in neighborhood composition and environmental quality. Thus, the clustering output is not merely a numerical partition induced by the mixture fit; it yields a meaningful summary of heterogeneity in the Boston housing data.

4.3. Prediction for new observations

Cluster assignment for new observations is performed through the same prediction interface. The output is a label in the space of the representative training clusters, obtained by aggregating posterior predictive co-clustering evidence at the cluster level.

```
z_test <- predict( fit_clust, newdata = test_dat, type    = "label")
```

Figure 7 displays the distribution of predicted cluster assignments across the test sample, produced by the package `plot()` method applied to the test labels object.

```
summary(z_test)$cluster_profiles

  cluster  n medv_mean  medv_sd lstat_mean lstat_sd
1      C4 53  21.57736 6.867192  13.415660 5.602698
2      C2 24  18.69583 5.343951  14.474583 7.229240
3      C5 18  35.15556 9.010988   5.679444 1.503887
```

```
4      C1  7  11.72857 2.293261  24.204286 6.162083
   rm_mean     rm_sd  nox_mean     nox_sd
1 6.225340 0.6417186 0.5775660 0.11813723
2 5.976583 0.4887950 0.5618333 0.12035912
3 7.260167 0.6143898 0.4957222 0.09672649
4 5.763571 0.5772723 0.6538571 0.09705227
```

```
plot(z_test, type = "sizes")
```

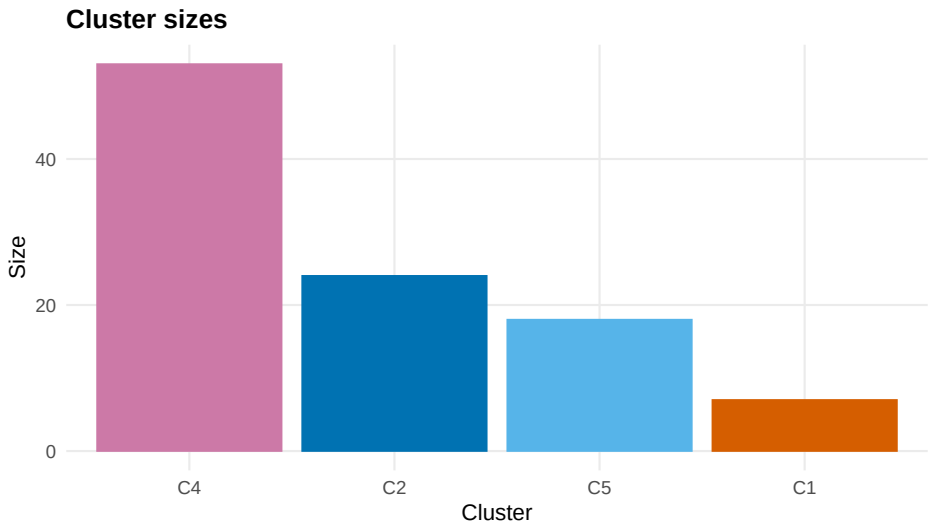


Figure 7: Bar plot of predicted cluster assignments for the held-out Boston test sample (the 20% of `medv` observations excluded from training), returned by `plot(predict(fit_clust,newdata=test_dat,type="label"),type="sizes")`. Each bar corresponds to one representative training cluster from Figure 6; bar heights give the number of test tracts assigned to that cluster.

The expected characteristics of the clusters' profiles are consistent with the interpretation of the representative partition. Cluster C5 contains the wealthiest tracts defined by a high mean house value, low average `lstat`, large mean number of rooms, and relatively low pollution. Cluster C1 represents the poorest neighborhoods with a low average house value, significant disadvantages in terms of neighborhood quality, low mean number of rooms, and a high level of pollution. Clusters C2 and C4 represent intermediate cases which show that multiple clusters were detected within the housing market rather than a bimodal distribution with poor and wealthy neighborhoods at its ends.

As expected, the vast majority of held-out tracts are classified into middle-value clusters (C4 and C2) with no tendency toward the low-end cluster. This is consistent with the Boston data, confirming that predictions are not unduly influenced by the outliers in the dataset. Tract assignments in the test set confirm the interpretation of the training-set partition: tracts belonging to more expensive clusters exhibit less neighborhood disadvantage, larger homes, and lower pollution, while tracts in poorer clusters exhibit the opposite characteristics.

This case study illustrates the workflow in which inference is summarized through a representative partition and cluster descriptions, and new observations are assigned to those clusters based on posterior predictive membership probabilities. The representative clusters also quantify the relationship between partition uncertainty and the marginal distributions of the response and covariates. The latent clusters align with the socioeconomic and environmental gradient, confirming that the nonparametric mixture recovers stable structure despite heterogeneity in the response. Causal comparisons are taken up in Section 5.

5. Data Analysis II: Causal Inference

This section describes the estimands from the two-arm spliced density regression. In contrast to the partition estimation of Section 4, the focus here is on mean and quantile differences for both unconditional and conditional causal estimands.

Data and estimands

To estimate unconditional and conditional causal estimands using the proposed approach, the `lalonde` dataset from the MatchIt R package is used for benchmarking. The outcome is the 1978 annual income level measured by `re78` ($), and the treatment indicator corresponds to program participation (variable `treat`). Covariates include age, years of schooling, race, marriage status, college degree status, and prior earnings `re74` and `re75`. Earnings typically exhibit considerable skewness, concentrate near zero, and have a pronounced upper tail, motivating the use of quantile and tail summaries rather than the mean alone.

The four estimands in Table 1 are computed following Section 2.5. Standardized estimands (ATE, QTE) are averaged across the empirical distribution of $\{\boldsymbol{x}_i\}_{i=1}^n$, while conditional estimands (CATE, CQTE) are evaluated at selected covariate profiles.

The data are first loaded and the design matrix constructed.

```
data("lalonde", package = "MatchIt")
app_lalonde_covars <- within(lalonde, {
  race <- factor(race, levels = c("white", "black", "hispan"))
  married <- factor(married, levels = c(0, 1), labels = c("no", "yes"))
  nodegree <- factor(nodegree, levels = c(0, 1), labels = c("no", "yes"))
})
app_lalonde_scale_vars <- c("age", "educ", "re74", "re75")
app_lalonde_scale_center <- vapply(
  app_lalonde_scale_vars,
  function(v) mean(app_lalonde_covars[[v]], na.rm = TRUE),
  numeric(1)
)
app_lalonde_scale_scale <- vapply(
  app_lalonde_scale_vars,
  function(v) stats::sd(app_lalonde_covars[[v]], na.rm = TRUE),
  numeric(1)
)
for (v in app_lalonde_scale_vars) {
  app_lalonde_covars[[paste0(v, "_z")]] <-
    (app_lalonde_covars[[v]] - app_lalonde_scale_center[[v]]) /
    app_lalonde_scale_scale[[v]]
}
app_lalonde_x_formula <- ~ age_z + educ_z + race + married + nodegree + re74_z + re75_z
app_lalonde_y <- (app_lalonde_covars$re78 + 0.5) / 1000
app_lalonde_A <- as.integer(app_lalonde_covars$treat)
app_lalonde_taus <- c(0.25, 0.50, 0.75, 0.90, 0.95)
app_lalonde_X <- stats::model.matrix(app_lalonde_x_formula, data = app_lalonde_covars
                                                        )[, -1, drop = FALSE]
```

Figure 8 provides full-sample descriptive views of observed 1978 earnings (`re78`) by treatment assignment. Both groups exhibit substantial mass near zero together with pronounced right-skewness, and the arm-specific distributions differ in ways that are not well summarized by a simple location shift. These features motivate modeling the full treatment-specific outcome distributions directly rather than relying solely on average effects.

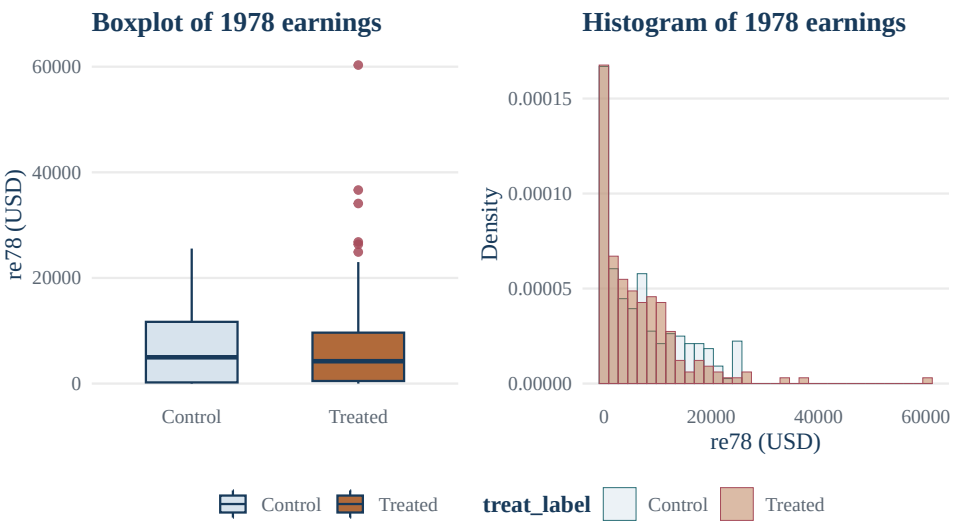


Figure 8: Full-sample distribution of 1978 annual earnings (`re78`, in US dollars) from the `MatchIt::lalonde` data set, stratified by treatment arm. Left panel: arm-specific boxplots with upper outliers highlighted. Right panel: overlaid density-scaled histograms for the control and treated arms.

5.1. Model and fit

Arm-specific conditional outcome distributions $F_a(\cdot \mid \boldsymbol{x})$ are modeled via the spliced bulk–tail construction described in Section 2.3:

$$f_a(y \mid \boldsymbol{x}) = \sum_j w_{aj}(\boldsymbol{x})\, k(y \mid \Theta_{aj})\, \mathbf{1}(y \le u) + \{1 - F_a(u \mid \boldsymbol{x})\}\, g(y \mid u, \sigma_a, \xi_a)\, \mathbf{1}(y > u).$$

The bulk kernel $k(\cdot)$ represents a Dirichlet process mixture regression with a Chinese restaurant process backbone, while the tail density $g(\cdot)$ follows the GPD distribution with positive scale $\sigma_a > 0$ and tail index ξ_a. In our model, the GPD tail index ξ_a regulates tail heaviness (e.g., if $\xi_a > 0$, it indicates a heavier tail). Our model mitigates poor finite-sample precision for extreme quantiles through nonparametric bulk and EVT-consistent tail modeling.

Here, a two-sample mixture is fitted with the gamma bulk kernel and the same MCMC configuration throughout. The aim is not to draw definitive conclusions but to demonstrate how the package produces standardized and conditional earnings contrasts under the spliced DPM-GPD framework.

```
fit <- dpmgpd.causal(y = app_lalonde_y, X = app_lalonde_X, treat = app_lalonde_A,
                     backend = "crp", kernel = "gamma", components = 10, mcmc = mcmc_fixed)
```

5.2. Posterior summaries

The fitted causal object defines two arm-specific outcome distributions from which all reported summaries are derived. The framework is therefore not limited to a single regression coefficient or scalar contrast; mean, quantile, and tail-sensitive causal estimands are all coherent functionals of the same fitted posterior distribution.

```
   total    ps    con    trt parallel_arms
1 336.35 32.98 199.02 104.28         FALSE
```

5.3. Standardized estimands: ATE and QTE

The overall treatment effect is summarized with ATE and QTE.

```
ate_fit <- ate(fit, interval = "hpd", level = 0.95)
summary(ate_fit)

ATE Summary
==================================================
Prediction points: 1
Conditional: NO | PS used: YES
Interval: hpd

Model specification:
  Backend (trt/con): crp / crp
  Kernel (trt/con): gamma / gamma
  GPD tail (trt/con): YES / YES

ATE estimates:
 estimate  lower upper
   -0.766 -2.992 1.358
```

The posterior ATE estimate is negative on the scaled earnings scale, but the corresponding interval includes zero. Thus, after averaging over the empirical covariate distribution, the analysis does not provide strong evidence of a uniform overall mean effect of the training program. Mean-based summaries alone are therefore insufficient to characterize the program effect in this application.

```
qte_fit <- qte(fit, probs = c(0.25, 0.50, 0.75), interval = "credible")
```

Figure 9 displays two standardized QTE views side by side from the spliced causal model: the effect curve and the treated-versus-control quantile curves with uncertainty intervals.

```
qte_effect_plot <- plot(qte_fit, type = "effect")
qte_arms_plot <- plot(qte_fit, type = "arms")
qte_effect_plot + qte_arms_plot + plot_layout(ncol = 2)
```

The QTEs remain negative at the specified quantile levels, indicating that the treatment distribution lies below the control at those points of the marginal distribution. The marginal quantile curves for both arms show that the deviations are not reducible to a location shift but reflect heterogeneity across the outcome distribution.

5.4. Conditional estimands for representative profiles

To characterize heterogeneity in the treatment effect, conditional estimands are computed at four representative covariate profiles spanning the observed covariate range. Profile values are expressed on the original data scale, using empirical quantiles for continuous variables and observed levels for categorical variables. Continuous predictors are standardized using training-set means and standard deviations before being passed to the model.

```
cate_fit <- cate(fit, newdata = xnew, interval = "hpd", show_progress = FALSE)
summary(cate_fit)
# plot(cate_fit, type = "effect") # Displays mean-based CATE plot by profile
```

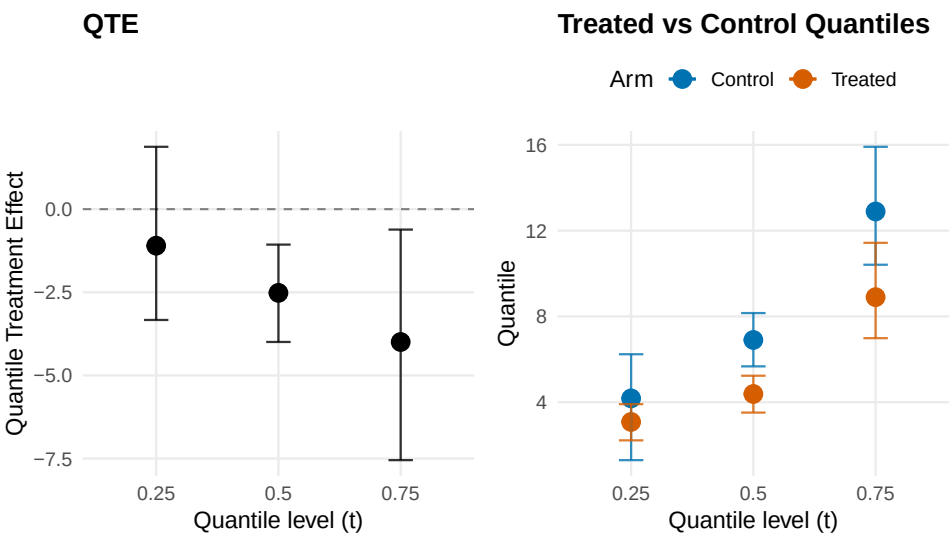


Figure 9: Standardized quantile treatment effect (QTE) summaries for Lalonde 1978 earnings, returned by `qte(fit,probs=c(0.25,0.50,0.75),interval="credible")`. Left panel: effect curve $Q_1^{\mathrm{m}}(\tau) - Q_0^{\mathrm{m}}(\tau)$ versus τ with pointwise 95% credible band. Right panel: standardized treated and control quantile curves $Q_1^{\mathrm{m}}(\tau)$ and $Q_0^{\mathrm{m}}(\tau)$ on the original earnings scale, with pointwise credible bands.

Table 2: Representative Lalonde participant profiles used as new-data inputs for the conditional causal contrasts in Figure 10. Columns (Profile 1–Profile 4) give the covariate values for one synthetic individual on the original data scale: `age` (years), `educ` (years of schooling), `race`, `married`, `nodegree`, and prior annual earnings `re74` and `re75` (US dollars).

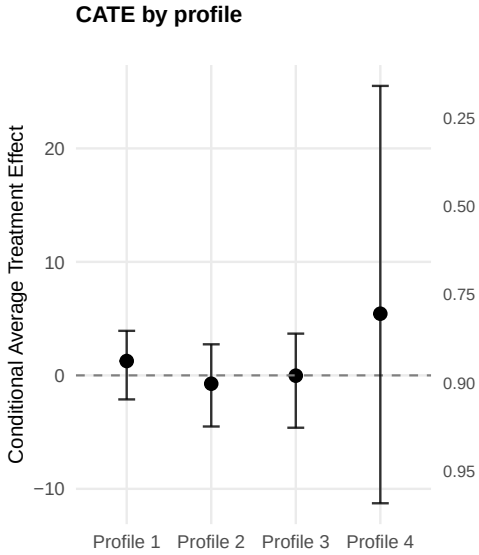
	Profile 1	Profile 2	Profile 3	Profile 4
age	20.0000	25.0000	32.0000	43.0000
educ	9.0000	11.0000	12.0000	13.0000
race	black	hispan	white	black
married	no	no	yes	yes
nodegree	yes	no	no	yes
re74	0.0000	47.4143	2705.7476	9383.2356
re75	0.0000	16.4710	1389.6486	4201.8870

The mean-based conditional average treatment effects are generally modest and uncertain. Although some profiles show positive point estimates, the corresponding intervals are broad and typically overlap zero. Thus, at the level of conditional means, the program does not appear to induce a strong and uniform shift in earnings across all participant types.

```
cqte_fit <- cqte(fit, probs = c(0.25, 0.50, 0.75, 0.90, 0.95),
                 newdata = xnew, interval = "credible")
# cqte_fit$fit_df # Displays the full CQTE posterior summary data frame
```

Figure 10 places the mean-based CATE plot beside a compact CQTE table so the profile-specific mean and quantile contrasts can be read together. The table entries report posterior estimates with 95% credible intervals in brackets.

The CQTE estimates reveal heterogeneity across both covariate profiles and quantile levels. At low and median quantile levels, the estimated program effect is close to zero or negative. Uncertainty about treatment effects is high at the upper quantiles, as upper-quantile contrasts rely on fewer observations and are more sensitive to the spliced tail. The analysis provides no evidence of positive program effects; any such effects, if present, appear confined to specific covariate profiles at upper quantiles.



CQTE with 95% credible intervals

	P1	P2	P3	P4
0.25	2.07 [−0.00, 3.55]	0.26 [−4.34, 3.39]	0.08 [−4.60, 3.42]	2.15 [−0.00, 3.66]
0.50	1.67 [−2.23, 4.12]	−2.02 [−5.52, 2.22]	−1.61 [−5.38, 1.24]	3.05 [−4.20, 13.28]
0.75	0.44 [−3.87, 5.38]	−2.07 [−6.01, 2.05]	−1.03 [−5.94, 3.15]	4.82 [−14.98, 32.88]
0.90	0.21 [−6.23, 7.18]	−1.24 [−6.93, 5.97]	0.50 [−5.67, 7.96]	8.71 [−30.39, 65.45]
0.95	0.79 [−8.46, 10.88]	−0.31 [−8.23, 11.01]	2.31 [−6.63, 14.03]	14.82 [−41.45, 104.81]

Figure 10: Conditional treatment-effect summaries for Lalonde 1978 earnings at the four representative covariate profiles (Profile 1–Profile 4) of Table 2. Left panel: profile-specific conditional average treatment effects $\mathrm{CATE}(x) = \mu_1(x) - \mu_0(x)$ with 95% HPD intervals, returned by `cate(fit,newdata=xnew,interval="hpd")`. Right panel: profile-specific conditional quantile treatment effects $\mathrm{CQTE}(\tau \mid x) = Q_1(\tau \mid x) - Q_0(\tau \mid x)$ at $\tau \in \{0.25, 0.50, 0.75, 0.90, 0.95\}$, returned by `cqte()`; each cell reports the posterior estimate with the 95% credible interval in brackets.

5.5. Results and interpretation

Two questions bear on the interpretation: the overall program effect on earnings, and heterogeneity of effects across participant profiles. The first yields no statistically compelling evidence — the average treatment effect is close to zero and the estimated QTE are negative at lower and mid-quantiles. The second yields equivocal findings: treatment effects are profile-specific, and some CQTE at upper quantiles show a positive effect confined to particular subpopulations.

This analysis underscores the value of quantile-based methods for heavy-tailed outcomes. A mean-only analysis provides little evidence of a program effect; the QTE, by contrast, reveals heterogeneity in the treatment effect across the outcome distribution.

6. Discussion

CausalMixGPD combines flexible bulk density estimation, optional EVT-based tail extrapolation, predictor-dependent clustering, and quantile- and density-level causal inference within one package. The same fitted model supports posterior summaries for densities, survival curves, analytical means (when they exist), restricted means, quantiles, and treatment contrasts. This is useful in settings where heavy tails and density-level effects are both of concern.

The package is considerably broader than the spliced examples emphasized in this article. Users may work with bulk-only or spliced models, multiple kernel families on the real line or on positive support, stick-breaking or CRP backends, and clustering or causal wrappers built upon the same predictive machinery. This shared infrastructure lets users move from exploratory subgroup analysis to tail-focused prediction and causal reporting without changing the underlying modeling language.

Several practical limitations warrant acknowledgment. First, computation can be demanding. Mixture models with covariate dependence, optional tail components, and arm-specific causal

fits are inherently MCMC-intensive, and costs escalate further with sensitivity analyses, longer chains, or richer predictor sets.

Second, model tuning requires attention. A spliced analysis inherits specification choices from both the DPM and peaks-over-threshold components, including kernel family, truncation level, threshold specification, and the number of exceedances available for tail learning. When overlap is weak or exceedances are sparse, posterior uncertainty can be substantial; for clustering, the posterior similarity matrix is often more informative than any single partition summary.

Third, the current scope is circumscribed. The causal implementation is restricted to binary treatments and continuous outcomes under standard consistency, ignorability, and overlap assumptions. Propensity-score augmentation can improve covariate adjustment in observational studies, but it is not a substitute for careful study design, overlap diagnostics, or sensitivity analysis for unmeasured confounding.

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A. Customization of Package and Additional Features

This appendix details the customizable features of CausalMixGPD. It covers argument-level customization for model fitting, posterior manipulation, and workflow-specific extensions.

A.1. Model fitting options and prior specifications

Besides the customizable features illustrated in Sections 2.1, 2.2, 2.5, and 3.1, this section lists arguments that allow kernel choice, latent structure, and prior customization alternative to default choices.

- **Kernel families and support** (): the argument `kernel` allows to choose among seven different kernel families, defined either with positive support or on the real line. The kernels with positive support are `"gamma"`, `"lognormal"`, `"invgauss"`, and `"amoroso"`; the kernels with real support are `"normal"`, `"laplace"`, and `"cauchy"`. The function `kernel_support_table()` reports the support, whether the kernel is compatible with GPD, and the supports of the covariates and backend. The function `get_kernel_registry()` returns the kernel registry with detailed information such as the parameter names, the default prior mode, and the density/CDF/quantile functions. For causal models, `kernel` can be either a single string or a length-two vector specifying different DPM kernels for the two arms. The kernel choice determines the support of the outcome variable and should be made based on the characteristics of the data.
- **Backend type and truncation rule** (): the argument `backend` determines the structure of the mixing coefficients. The value `sb` stands for the stick-breaking scheme while `crp` denotes the Chinese Restaurant Process latent allocation scheme. For causal models, `backend` can be either a single string or a length-two vector specifying different latent allocation schemes for the two arms. The argument `components` sets the maximum number of mixture components (defaulting to the sample size). The argument `epsilon` specifies a truncation threshold below which components are dropped and the smallest weight is readjusted. Both arguments can be customized separately for each causal arm.
- **Parameter specification and prior setting** (): the argument `param_specs` is a named list giving the parameters of the kernel or tail distribution. Each kernel/tail parameter supports three modes: `"fixed"`, `"dist"`, and `"link"`. With `"dist"`, the prior can be chosen as any distribution compatible with the parameter and supported by nimble, with

hyperparameters specified in the default registry. With "link", the kernel/tail parameter is related to the covariates through a linear regression using one of five link functions ("identity", "exp", "log", "softplus", or "power"); the link power is adjusted via link_power when the last option is selected. With "fixed", the parameter is fixed to the constant value specified in param_specs. For causal models, param_specs can be either a single named list or a length-two list specifying different parameter specifications for the two arms.

A.2. Fitting, prediction, and posterior output

This section gathers the arguments used after a model class has been chosen: sampler configuration, monitoring, prediction targets, interval summaries, and extraction utilities.

- **MCMC configuration** (): mcmc controls niter, nburnin, thin, nchains, and seed; causal models split this into mcmc_outcome and mcmc_ps. Runtime controls include show_progress, quiet, parallel_chains with workers, parallel_arms for arm-wise causal fitting, and timing. CRP-based backends additionally use z_update_every to tune the frequency of global label updates.
- **Monitoring and fitting interfaces** (): monitor="core" retains the main structural parameters, whereas "full" retains all nodes. monitor_latent=TRUE stores allocation labels $z_1, \ldots, z_n$, and monitor_v=TRUE stores stick-breaking fractions for SB models. Models can be supplied through a formula interface (formula, data) or directly through y and X; causal fits additionally use treat. The bundle() plus mcmc() workflow separates model construction from sampling when the generated nimble object needs to be inspected or modified first.
- **Prediction types and intervals** (): predict() supports type="density", "survival", "quantile", "mean", "rmean", "median", "sample", and "fit". These return posterior summaries of $f(y \mid x)$, $S(y \mid x)$, $Q(\tau \mid x)$, $\mathrm{E}(Y \mid x)$, the restricted mean $\mathrm{E}[\min(Y, c) \mid x]$ via cutoff and nsim_mean, the median (as a special case of the quantile method), posterior predictive draws via nsim, or retained draw-level fitted values for custom checks. Interval summaries are controlled by interval="credible", "hpd", or NULL, together with level.
- **Summary and extraction methods** (): summary() reports posterior means, standard deviations, quantiles, and Effective Sample Size(ESS). params() reshapes posterior means into component- and coefficient-level layouts; fitted() and residuals() provide observation-level predictive summaries.
- **Visualization and extraction** (): plot() for one-arm fits covers standard MCMC diagnostics, including "traceplot", "density", "histogram", "running", "autocorrelation", "crosscorrelation", "Rhat"/"grb", "effective", "geweke", "caterpillar", "pairs", and family="auto". Companion extraction tools include summary(), params(), fitted(), residuals(), and ess_summary().

A.3. Workflow-specific extensions

This section records controls specific to the clustering and causal workflows in Sections 3.3, 3.4, 4, and 5. Representative outputs appear in Figures 6, 7, 9, and 10, with the latter combining the CATE plot and CQTE table.

- **Causal estimands** (): beyond ate(), qte(), cate(), and cqte(), the package provides att() for effects averaged over the treated covariate distribution, qtt() for quantile effects on the treated, and ate_rmean() for restricted-mean contrasts. The ate() and cate()

families also accept `type="mean"` or `"rmean"` with a corresponding `cutoff`. All effect objects support `summary()`, `print()`, and `plot()` S3 functions. Causal plots route by arm and offer `type="effect"`, `"arms"`, or `"both"`; QTE plots additionally use `facet_by`; cluster plots provide PSM heatmaps plus label summaries via `type="sizes"`, `"certainty"`, and `"summary"`. All plots return ggplot2 objects and can be converted to plotly.

- **Propensity score customization** (): observational causal fits can estimate propensity scores with `PS="logit"`, `"probit"`, `"naive"`, or `FALSE`. The estimated score is appended to the outcome model on either the logit scale (`ps_scale="logit"`) or probability scale (`"prob"`); posterior summaries use `ps_summary="mean"` or `"median"`. Numerical stability is controlled through `ps_clamp`, coefficient shrinkage through `ps_prior`, and the two-stage fitting scheme through separate `mcmc_ps` and `mcmc_outcome` lists.
- **Clustering workflow and prediction types** (): `dpmix.cluster()` and `dpmgpd.cluster()` accept `type="weights"`, `"param"`, or `"both"` to place predictor dependence in the mixture weights, kernel parameters, or both. Their `predict()` methods return either a posterior similarity matrix (`type="psm"`) or representative labels under the Dahl criterion (`type="label"`), including assignments for `newdata`. The associated `summary()` method reports cluster sizes and within-cluster covariate/response profiles.

A.4. Utilities and Computing Support

The last appendix section includes a selection of helper functions and resources that accompany the main modeling process.

- **Density and quantile utilities** (): the package provides the complete R-style `d/p/q/r` functions for GPD, Inverse Gaussian, Amoroso Distribution with every supported kernel mixture, and every spliced kernel mixture and GPD distribution. Typical examples include `dgpd()`, `dgammamix()`, and `dgammamixgpd` which are density functions of the GPD, gamma mixture, and spliced gamma mixture with GPD tail, respectively that can be used directly in modeling and posterior manipulation in Nimble. They are particularly useful for prior checking, generating simulations, benchmarking, and other purposes requiring density and quantile manipulations.
- **Simulators and bundled datasets** (): functions `sim_bulk_tail()`, `sim_causal_qte()`, and `sim_survival_tail()` provide data generation for reproducible benchmarks for bulk and tail and causal analyses; random number generation seeds are supplied by the `seed` argument. The package comes bundled with twelve datasets with unconditional/conditional/causal, positive/real-line, and tail/no-tail designs, each featuring outcomes, covariates, treatment indicators where relevant, metadata, and the true values of interest.
- **Computational considerations** (): large-scale analyses allow the use of `parallel_chains` with `workers`, `parallel_arms` for causal fits, `chunk_size` for predicting in parallel chunks, and `ndraws_pred` for sampling from the posterior predictive distribution of parameters. The `show_progress` and `quiet` arguments can be used to control progress reporting and compiler verbosity, respectively, while `timing=TRUE` saves time spent on each fit for benchmarking purposes.

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